

Question 1

PERSONAL ANALYSIS:

I understand that we need to develop a program that will help the user to keep records of each event that happens to their bank account, whether a withdrawal, a deposit or an automatic deduction.

GROUP ANALYSIS:

We need to use arithmetic operations and variables to record each event. We shall display the current account balance after each event. We may also use loops to allow any type of event to be recorded multiple times.

FINAL ANALYSIS:

Now, I conclude to use arithmetic operators in the system to carry out some arithmetic operations and loops to allow multiple entries. In addition, I need to include conditions such that the system will allow the user to make a choice of what event is going to be recorded. This is more flexible and makes a program more reliable.

Pseudocode

```
Initialize balance to 100
SET status = True
PRINT "Select the event"
WHILE status is True DO
    Display options:
        1: New Deposit
        2: Withdrawal
        3: Record Interest
        4: Monthly Fee
        5: Exit
    GET user input and store in choice
    IF choice is '1' THEN
        PROMPT "How much is the new deposit:"
        GET input and name it amount
        CALCULATE balance += amount
    ELSE IF choice is '2' THEN
        PROMPT "How much did you withdraw:"
        GET input and set it to amount
        CALCULATE balance -= amount
    ELSE IF choice is '3' THEN
        CALCULATE balance = balance - (balance * 5 / 100)
    ELSE IF choice is '4' THEN
        CALCULATE balance = balance - 10
    ELSE IF choice is '5' THEN
        PRINT "The program closed"
        SET status to False
    ELSE
        Display "Invalid input"
    Display "The current balance is: ", balance
END WHILE
```

Question 2

PERSONAL ANALYSIS:

All we are expected to do is to create a program that will allow the user to calculate the area and the circumference of the compound given the diameter of that compound.

GROUP ANALYSIS:

We need to find the total area and the circumference of the garden of the compound. We will use the python math module to declare the pie variable. We may also use other simple arithmetic operations and variables.

FINAL ANALYSIS:

We need a pseudocode of a program that finds the area and circumference of the 7.5 metres diameter long circular compound using the value of pie extracted from the python math library.

PSEUDOCODE:

BEGIN

 IMPORT math module

 SET diameter = 7.5

 CALCULATE radius = diameter ÷ 2

 SET area = math.pi * radius ** 2

 SET circumference = math.pi * diameter

 SET rounded_area = ROUND(area, 2)

 SET rounded_circumference = ROUND(circumference, 2)

 PRINT "The area of the garden will be", rounded_area, "square meters and the circumference is", rounded_circumference, "meters."

END

Question 3

PERSONAL ANALYSIS

We are expected to create a program for measuring the ingredients of the cookies that will fit 20 people.

GROUP ANALYSIS:

We will find the constant 20/8 and multiply it by the other resources' quantities. We need to change the initial temperature of 180 °C to Fahrenheit degrees to make sure the temperature of the new oven stays the same as the old one's. We will also use arithmetic operations.

FINAL ANALYSIS

SET original_people TO 8

SET new_people TO 20

SET flour_per_8 TO 300

SET eggs_per_8 TO 4

SET baking_time TO 45

SET temp_celsius TO 180

COMPUTE scale_factor AS new_people ÷ original_people

COMPUTE new_flour = flour_per_8 * scale_factor

COMPUTE new_eggs = eggs_per_8 * scale_factor

CONVERT temp_celsius TO Fahrenheit:

SET temp_fahrenheit TO (temp_celsius * 9 / 5) + 32

DISPLAY "New Recipe for", new_people, "people"

DISPLAY "Flour:", new_flour, "grams"

DISPLAY "Eggs:", new_eggs

DISPLAY "Baking Time:", baking_time, "minutes"

DISPLAY "Oven Temperature:", temp_fahrenheit, "°F"

Question 4

PERSONAL ANALYSIS:

Clara was estimating book sales and restocks instead of recording exact numbers, causing inconsistencies. She starts with 200 books on Monday. To automate and avoid confusion, the program will apply fixed values: sell 30 and restock 25 from Monday to Friday, sell 45 on Saturday with no restock, and make no changes on Sunday. The goal is to calculate accurate daily inventory and the final count on Sunday.

GROUP ANALYSIS:

The group has agreed to develop a bookshop tracker by first collecting the daily number of books sold and then calculating the total at the end of each week. To achieve this, they will create an actual tracking system that utilizes loops for repeated actions.

FINAL ANALYSIS:

SET inventory TO 200

SET week TO ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"]

FOR each day IN week DO

 DISPLAY day and starting inventory

 IF day is Monday to Friday THEN

 SUBTRACT 30 from inventory

 ADD 25 to inventory

 ELSE IF day is Saturday, THEN

 SUBTRACT 45 from inventory

 ELSE IF day is Sunday THEN

 DO NOTHING

 DISPLAY day and ending inventory

END FOR

DISPLAY final inventory on Sunday night

Question 5

PERSONAL ANALYSIS:

The goal is to simulate daily energy balances from Monday to Sunday, following the same rhythm each day and ensuring Sunday ends with no cycles applied.

GROUP ANALYSIS:

Jeffrey's Energy Bank starts at 50 units on Monday, and each day from Monday to Saturday, the energy changes through 3 cycles: dawn adds 8 units, noon deducts 3 units, and dusk adds 5 units. But on Sunday, no changes happen after dusk.

FINAL ANALYSIS:

SET energy TO 50

SET week TO ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"]

FOR each day IN week DO

 DISPLAY day and starting energy

 IF day is not Sunday THEN

 CALCULATE energy += 8

 CALCULATE from energy -= 3

 CALCULATE energy += 5

 DISPLAY day and ending energy

END FOR

DISPLAY final energy balance at the end of Sunday